

will start with super-wide channels of 160MHz of 2.5GHz spectrum, which is already serviced by half of Sprint's towers in the top 100 US markets. Higher speeds will be achieved using Massive MIMO antennae that can communicate with devices on multiple frequencies at once; the antennae can be used now for faster 4G LTE service and then software-switched to 5G in 2019.

Interestingly, Sprint CEO Marcelo Claure claimed that Sprint will have the first mobile 5G network in the US and potentially in the world.

Soon every gadget will have AI inside

Betting big on artificial intelligence, Arm has announced a pair of chip designs for artificial intelligence, called Project Trillium. One is named the Arm Machine Learning processor, which can be used for any AI task from translation to facial recognition. The other is the Arm Object Detection processor, which finds interesting bits in images (like faces) and passes them onto the machine learning processor for processes like facial recognition.

Jem Davies, Arm VP of machine learning, told *Quartz* that these processor designs can be scaled up—meaning these could be used in Internet of Things (IoT) devices or modified and enlarged for use in datacentres.



Holy chip (Reuters/Toby Melville)

Amazon, Google, Apple, Intel, Qualcomm and a host of startups have all started designing their own AI-specific chips, signalling that the best software and hardware engineers are looking towards specialty chips to enable AI functions. Arm's entrance onto the scene can be taken as a sign that the hardware has arrived. It's no longer niche or special for device makers to embed an AI-specific chip in their gadgets, because now they can purchase this design from Arm.

Arm expects most of the growth for this technology to come from IoT devices and Internet-connected cameras.

Qualcomm chip enabling 5G launches in 2019

Qualcomm has announced that its Snapdragon X50 5G New Radio (NR) mobile modem chipset was chosen by 19 global operators for upcoming mobile 5G trials. The chip giant also received pledges from more than a dozen OEMs to use the



Qualcomm Snapdragon X50 5G New Radio chip is designed to enable field trials and early deployments of 5G networks globally

component for commercial 5G device launches due in 2019.

"2018 will be a big year for Qualcomm Technologies and the mobile industry overall as we work to execute on the agreed upon 5G NR specification," said Qualcomm president Cristiano Amon. "These trials demonstrate how we are working with global operators to fuel growth and innovation in the mobile industry and make 5G a reality by supporting a large number of expected commercial launches in 2019."

Qualcomm had unveiled the Snapdragon X50 in 2016, with plans to make it commercially available during the first half of 2019. It was designed to enable field trials and early deployments of 5G networks globally, allowing download speeds of around 5Gbps and supporting operation in the mmWave spectrum at the 28GHz band, with 800MHz bandwidth, along with multiple-input multiple-output (MIMO), adaptive beamforming and beam-tracking technologies.

The modem can be used for 4G and 5G mobile broadband, along with fixed-wireless broadband devices, when paired with a Qualcomm Snapdragon platform and its gigabit-speed 4G modem.

Intel launches chip for computing at the edge

As we develop increasingly sophisticated technologies like self-driving cars and industrial Internet of things sensors, we need to move computing to the edge. Essentially this means that instead of sending data to the cloud, processing must be done right on the device itself because even a little bit of latency is too much.



Intel's latest chip, Xeon D-2100 processor